

Who Really Talks About Cybersecurity on Bluesky?

COMP3020 Social Web Analytics — Group 32

COMP3020 — Social Web Analytics

Group Number: 32

Group Members and Contributions

Group Member	Student ID	Contribution
Yousuf	20397249	33.34%
Seemrah Panwala	22175168	33.33%
Rishikanth	22170637	33.33%
Total		100%

Before submission, replace abbreviated names with full names and confirm these contribution percentages reflect the group's agreed allocation.

2. Analysis

2.1 Data Collection

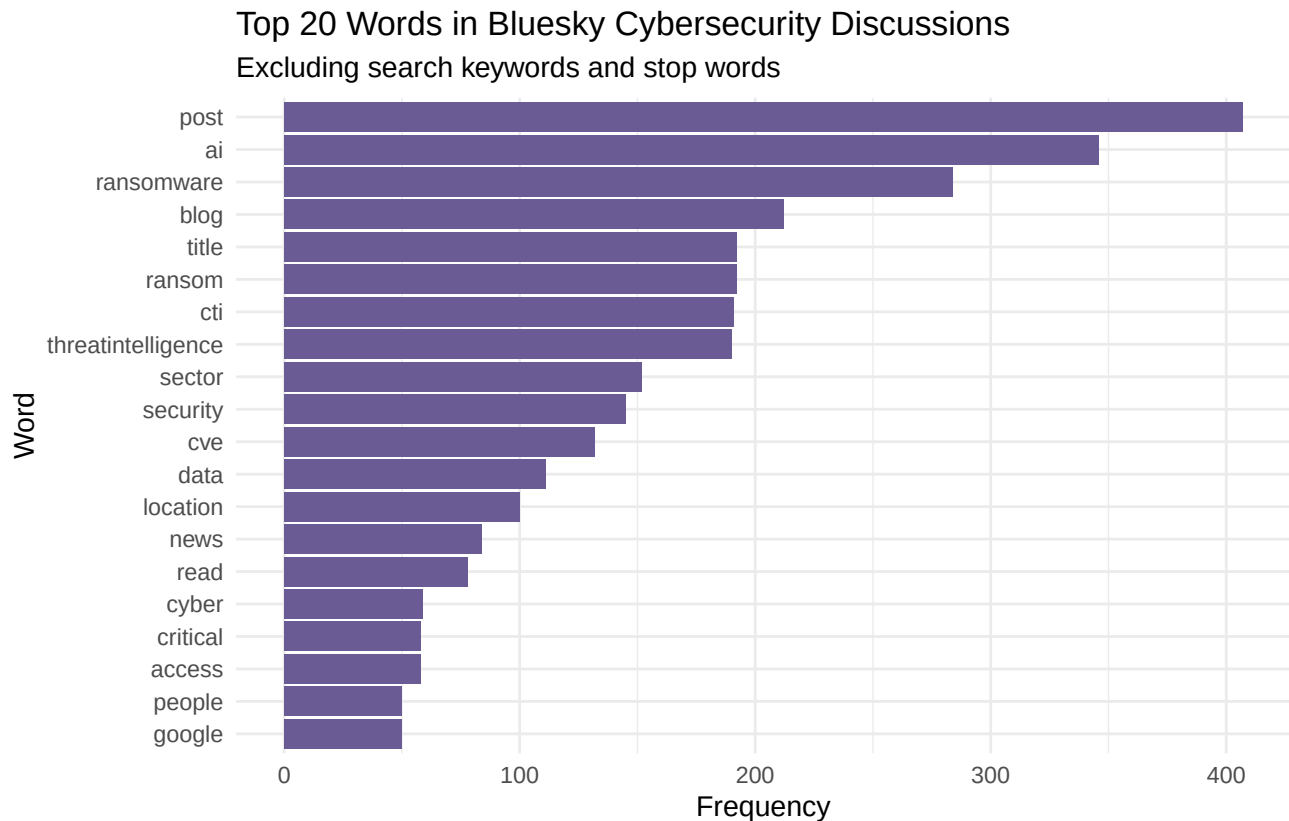
2.2 Hypothesis Testing

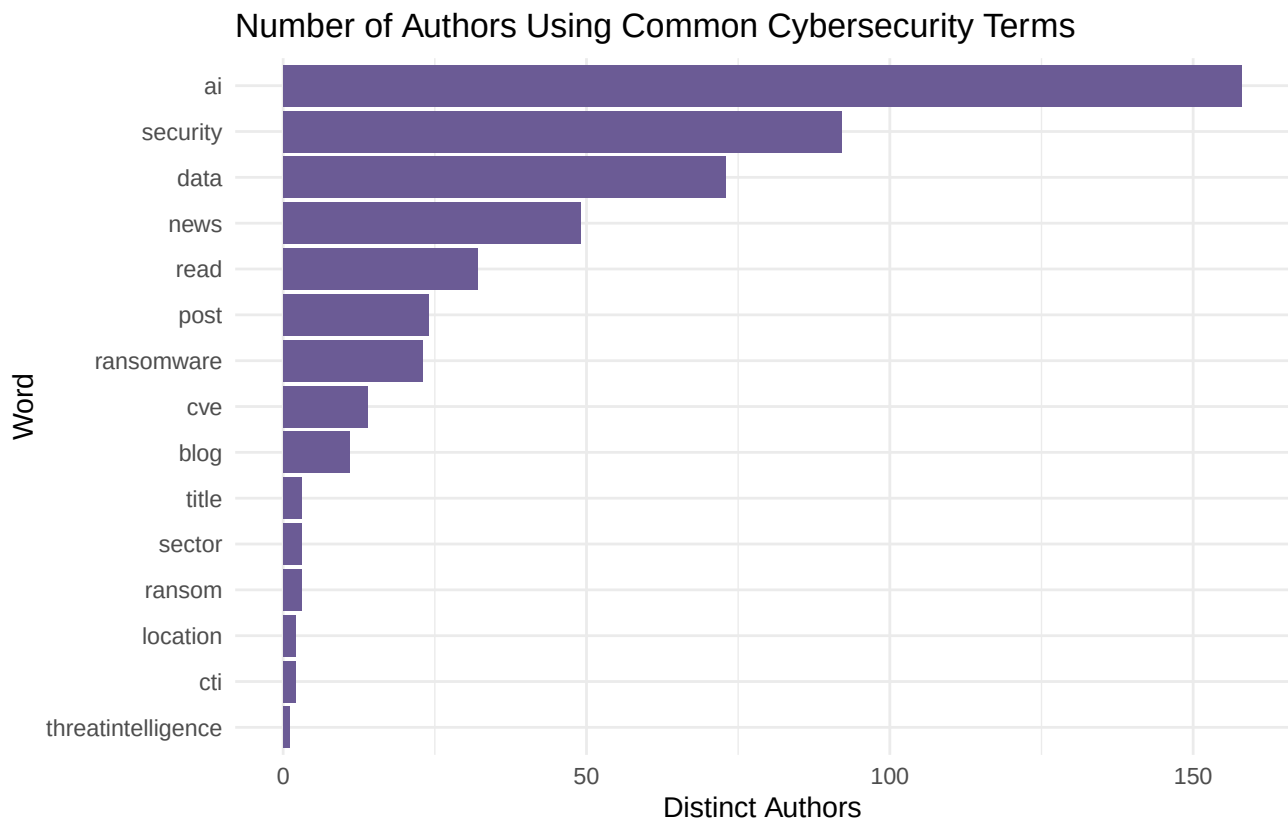
2.3 Text/Content Analysis and Visualisation

Research question: What are the most frequently discussed topics in cybersecurity-related posts on Bluesky?

This section analyses the text of cybersecurity-related Bluesky posts to identify frequently occurring words and emerging discussion themes.

```
custom_stop <- tibble(  
  word = c("cybersecurity", "infosec",  
           "https", "http", "www", "amp")  
)  
  
words <- posts %>%  
  select(id, text) %>%  
  filter(!is.na(text), text != "") %>%  
  mutate(text = str_remove_all(text, "https?://\\S+")) %>%  
  unnest_tokens(word, text) %>%  
  anti_join(stop_words, by = "word") %>%  
  anti_join(custom_stop, by = "word") %>%  
  filter(!str_detect(word, "^[0-9]+$"))
```





Interpretation of Text Analysis

The most frequent terms include AI, ransomware, threat intelligence and vulnerabilities. The author-level comparison adds an important distinction: AI was used by approximately 158 distinct authors, whereas ransomware appeared across approximately 23 authors, and terms such as “CTI” and “threatintelligence” were concentrated among one or two authors. These counts indicate that high term frequency does not necessarily represent broad participation. Repeated words such as “post”, “blog” and “title” may reflect structured news-sharing or templated content, but vocabulary alone cannot establish whether an account is automated.

Limitation: Keyword-based sampling, short posts, repeated feeds and imperfect language tags may affect the vocabulary observed. Word counts do not establish sentiment or context.

2.4 Clustering Analysis

Research question: What groups of similar topics can be identified in cybersecurity-related Bluesky posts?

```
custom_stop <- tibble(
  word = c("cybersecurity", "infosec",
           "https", "http", "www", "amp")
)

cluster_words <- posts %>%
  select(id, text) %>%
  filter(!is.na(text), text != "") %>%
  mutate(text = str_remove_all(text, "https?://\\S+")) %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word") %>%
```

```

anti_join(custom_stop, by = "word") %>%
filter(!str_detect(word, "[0-9]+$"))

tfidf_data <- cluster_words %>%
  count(id, word) %>%
  bind_tf_idf(word, id, n)

# Keep the 500 most frequent words to reduce noise
top_terms <- cluster_words %>%
  count(word, sort = TRUE) %>%
  slice_head(n = 500) %>%
  pull(word)

# Convert TF-IDF into a document-term matrix
tfidf_matrix <- tfidf_data %>%
  filter(word %in% top_terms) %>%
  select(id, word, tf_idf) %>%
  pivot_wider(
    names_from = word,
    values_from = tf_idf,
    values_fill = 0
  )

# Keep post IDs separately
post_ids <- tfidf_matrix$id

# Convert to numeric matrix
X <- as.matrix(tfidf_matrix %>% select(-id))

# Remove documents with no retained terms
keep <- rowSums(X^2) > 0
X <- X[keep, , drop = FALSE]
post_ids <- post_ids[keep]

# Normalise each document vector
X <- X / sqrt(rowSums(X^2))

set.seed(123)

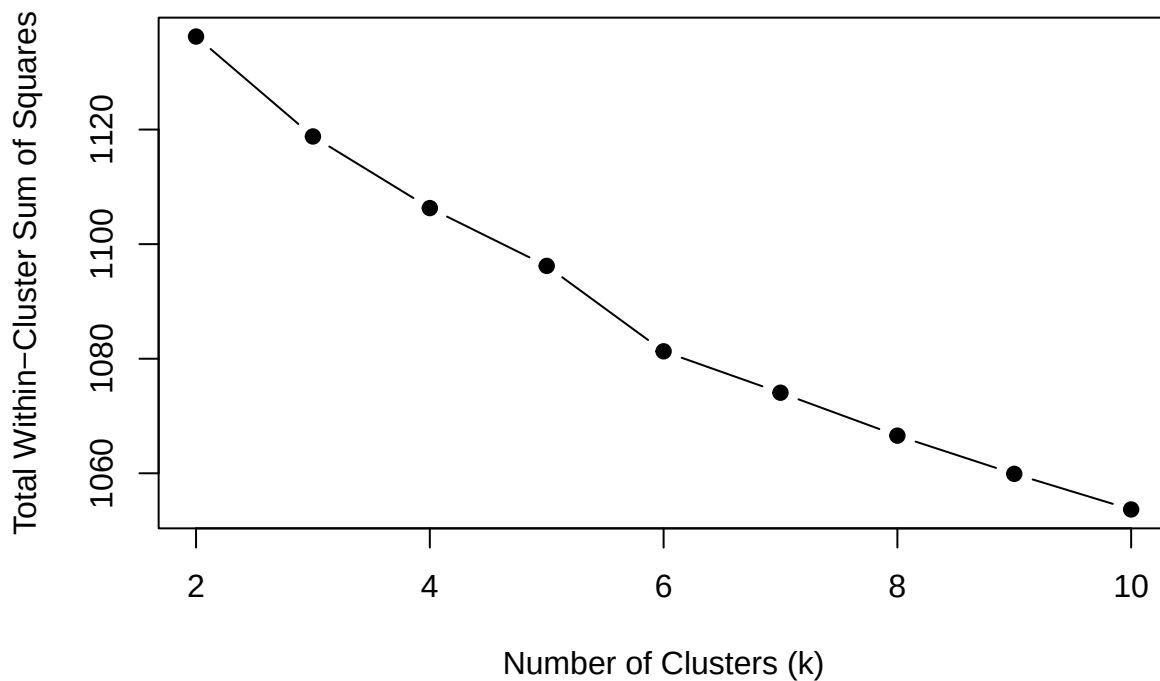
# Test different numbers of clusters
k_values <- 2:10

wss <- sapply(k_values, function(k) {
  kmeans(X, centers = k, nstart = 10, iter.max = 100)$tot.withinss
})

# Elbow plot
plot(
  k_values, wss,
  type = "b",
  pch = 19,
  xlab = "Number of Clusters (k)",
  ylab = "Total Within-Cluster Sum of Squares",
  main = "Elbow Method for Selecting k"
)

```

Elbow Method for Selecting k



```
set.seed(123)

k <- 6

km <- kmeans(
  X,
  centers = k,
  nstart = 25,
  iter.max = 100
)

cluster_assignments <- tibble(
  id = post_ids,
  cluster = factor(km$cluster)
)

# Check the number of posts in each cluster

cluster_terms <- tfidf_data %>%
  inner_join(cluster_assignments, by = "id") %>%
  group_by(cluster, word) %>%
  summarise(total_tfidf = sum(tf_idf), .groups = "drop") %>%
  left_join(cluster_assignments %>% count(cluster, name = "cluster_size"), by = "cluster") %>%
  mutate(mean_tfidf = total_tfidf / cluster_size) %>%
  group_by(cluster) %>%
  slice_max(mean_tfidf, n = 10) %>%
  arrange(cluster, desc(mean_tfidf))
```

The elbow curve begins to flatten around $k = 6$, although the bend is not sharply defined. We therefore selected six clusters as an exploratory choice rather than a uniquely optimal solution.

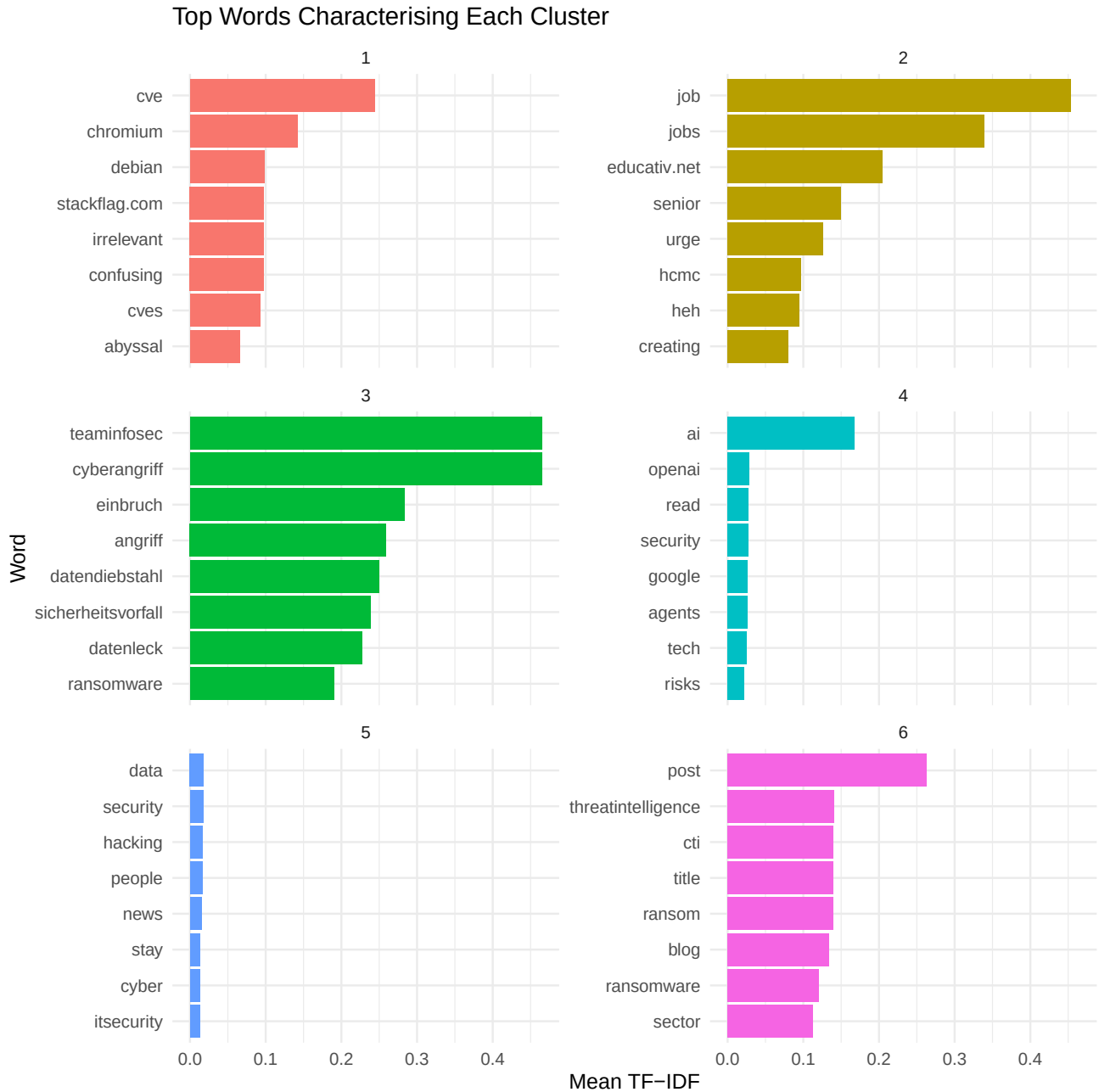
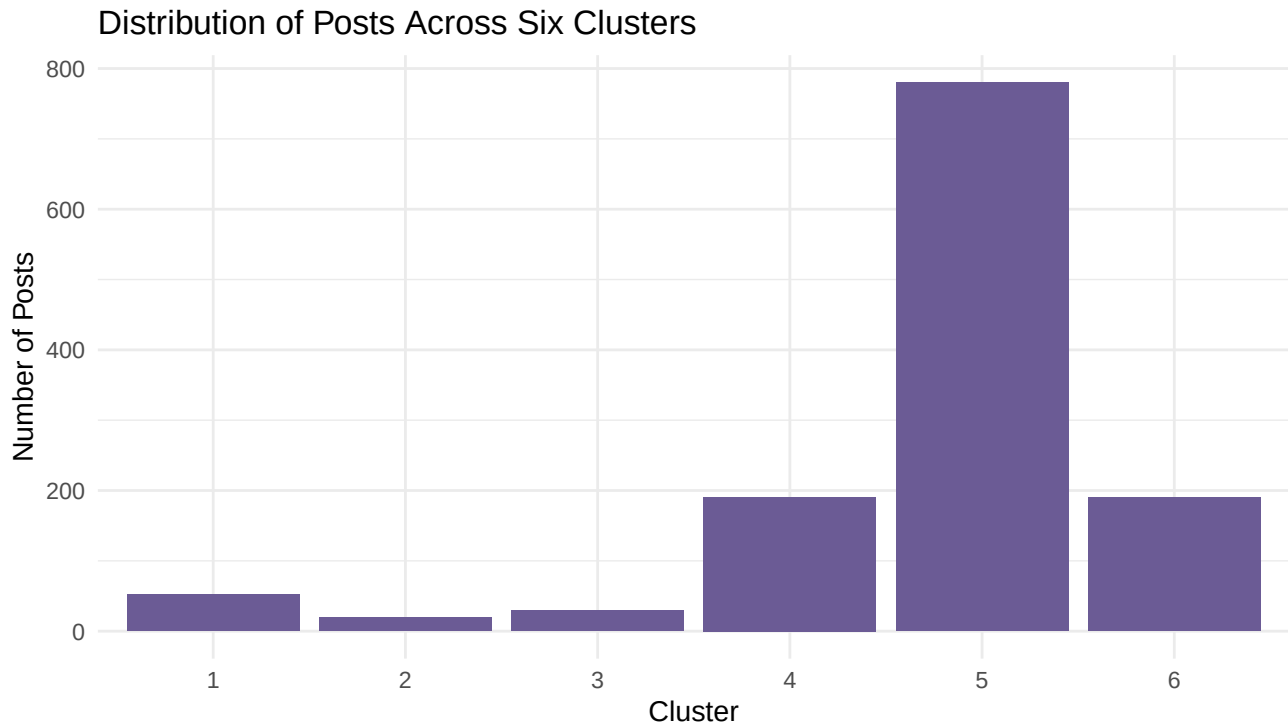


Figure 1: Highest average TF-IDF terms within each of the six clusters.



Looking at the distribution, Cluster 5 took up the lion’s share with 780 posts—roughly 62% of the clustered posts. Clusters 4 and 6 sat comfortably in the middle at around 190 posts each, with Cluster 6 heavily leaning into structured cybersecurity news and threat-intelligence terminology.

Meanwhile, Clusters 1, 2, and 3 were much smaller, but they packed a punch with highly specific vocabulary.

This contrast in cluster sizes makes a lot of sense: most people are sharing broad, overlapping cybersecurity talk, while smaller, dedicated groups naturally stick to far more distinctive, specialized language.

To make sense of the data, we grouped the posts into six clusters based on their TF-IDF profiles. While the elbow method gave us a bit of a faint signal rather than a sharp bend, six clusters gave us a really useful structure to work with.

Naturally, the data had its own story to tell. About 62% of clustered posts (780 in total) tumbled into Cluster 5, which naturally absorbed general, everyday cybersecurity talk. But when we looked past that main cluster, the remaining groups really shone—highlighting specific discussions around ransomware, job hunting, and technical vulnerabilities.

Cluster 6 gave us a clear view of bot-like or templated ransomware alerts, while Cluster 3 unexpectedly caught a cluster of German posts that slipped through our English filter. That was a great reminder of how real-world data constantly tests our assumptions! Ultimately, clustering proved its worth by revealing these distinct sub-communities that standard frequency counts would have completely overlooked.

2.5 Network Analysis

Research question. *In the follow network around the most-followed active author in the cybersecurity discussion, which accounts are most central, what kind of accounts are they, and does this network belong to the cybersecurity community?*

Why a follow network rather than a mention network

The project guide suggested a weighted mention network, in which an edge $A \rightarrow B$ means that A mentioned B. **This section deliberately deviates from that design**, because our data cannot support a mention network.

Only 40 of the 1,293 cleaned posts (3.1%) contain a mention, and they contain 52 mentions in total. A mention network would therefore have at most 52 edges spread across a few dozen posts. It would be too small and fragmented to measure centrality in a meaningful way. Bluesky also stores mentions as account IDs (DIDs) rather than handles, so the cleaned data keeps only the mention count. We therefore follow the Module 7 seed-user method and analyse a **follow network**, which the API returns directly. The two networks answer different questions. A mention network shows who talks to whom in cybersecurity posts. A follow network shows whose content an account has chosen to receive. All centrality results below are therefore about follow relationships, not about mention-based conversation.

Building the network

The seed is sifu.tweety.fish, the author with the most followers among authors who posted about the topic at least three times. We collected the 59 accounts the seed follows (its *friends*) and then the accounts each friend follows. 0 friend retrievals failed. All relationships come from the Bluesky API, so none are inferred by us.

- **Nodes** are Bluesky accounts.
- **Edges** are follow relationships. An edge $A \rightarrow B$ means “A follows B”.
- The graph is **directed** because following is one-way: A can follow B without B following A.
- The graph is **unweighted** because a follow either exists or does not. Unlike a mention, it has no count attached to it.

```
edges <- as.data.frame(edges_raw, stringsAsFactors = FALSE)
g <- graph_from_data_frame(edges, directed = TRUE)

# Role of each account in the sample
V(g)$role <- case_when(
  V(g)$name == seed_user_handle ~ "Seed",
  V(g)$name %in% friends_handles ~ "Friend (followed by seed)",
  TRUE ~ "Friend of friend"
)

# Link to the post data: did this account post about cybersecurity in our sample?
authors <- setdiff(unique(posts$author), "handle.invalid") # unresolved handle, not a real account
V(g)$cyber_author <- V(g)$name %in% authors
```

Network properties

```
comp_weak <- components(g, mode = "weak")
comp_strong <- components(g, mode = "strong")
tot_deg <- degree(g, mode = "all")
out_deg <- degree(g, mode = "out")
in_deg <- degree(g, mode = "in")

# Largest weakly connected component
gc <- induced_subgraph(g, comp_weak$membership == which.max(comp_weak$csizes))
```

The network has 3,201 accounts and 3,619 follow edges. Its density is 0.00035, which means only a tiny fraction of the possible connections exist. This is typical of social networks: each account follows only a handful of the thousands of others in the sample. Ignoring direction, the network forms a single weakly connected component, so the largest weakly connected component contains all 3,201 accounts. This is expected from the collection method, because every account was reached from the seed. Once direction is taken into account, the network is far less connected: there are 3,176 strongly connected components, and the largest has only 24 accounts. Only 1.4% of follows are returned, so attention in this network flows mostly one way.

Table 2: Properties of the follow network

Property	Value
Nodes (accounts)	3,201
Edges (follows)	3,619
Density	0.00035
Weakly connected components	1
Largest weakly connected component (nodes)	3,201
Strongly connected components	3,176
Largest strongly connected component (nodes)	24
Reciprocity (share of follows returned)	0.014
Mean total degree	2.26
Diameter (longest shortest path, directed)	4

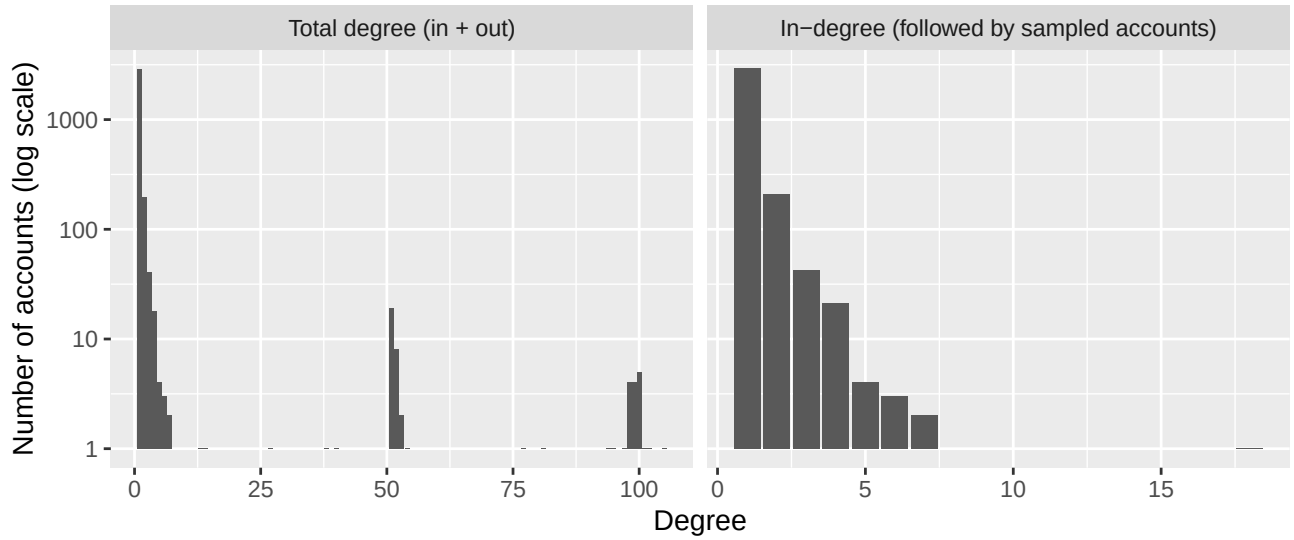


Figure 2: Degree distributions of the follow network: total degree (left) and in-degree (right). Count axis on a log scale

Table 3: Top 10 accounts by in-degree

account	role	in-degree	betweenness	posted?
sifu.tweety.fish	Seed	18	69,077	TRUE
regimecpa.bsky.social	Friend	7	6,593	FALSE
philpax.me	Friend of friend	7	0	FALSE
danioloc.xyz	Friend of friend	6	0	FALSE
segyges.bsky.social	Friend of friend	6	0	FALSE
theophite.bsky.social	Friend of friend	6	0	FALSE
elucidating.extradimensional.space	Friend of friend	5	0	FALSE
pfrazee.com	Friend of friend	5	0	FALSE
pixelatedboat.bsky.social	Friend of friend	5	0	FALSE
sky.skymarchini.net	Friend of friend	5	0	FALSE

Both distributions are highly skewed. In the **total-degree** distribution, 2,880 accounts (90%) have a degree of 1. The highest total degrees all belong to the seed and its friends (median 52, maximum 105). Their high degree mainly reflects the follow lists we collected, because only the seed and its friends have outgoing edges. The API returned between 1 and 99 follows per friend (median 50). The **in-degree** distribution is not affected by this collection cap, so it is the better measure of popularity. 2,917 accounts (91%) are followed by only one sampled account, while a few are followed by many. This long-tailed pattern means a small number of accounts act as hubs that many of the seed’s friends choose to follow.

Centrality

We use two centrality measures that answer different questions:

- **In-degree:** how many sampled accounts follow this account. A high in-degree means the account is popular or seen as worth following within the seed’s community.
- **Betweenness:** how many shortest paths between other accounts pass through this account. A high betweenness means the account is a *bridge* that connects parts of the network that would otherwise be far apart.

```
cent <- tibble(
  account = V(g)$name,
  role    = V(g)$role,
  in_degree = in_deg,
  betweenness = round(betweenness(g, directed = TRUE)),
  posted    = V(g)$cyber_author
)

top_in <- cent %>% arrange(desc(in_degree), desc(betweenness), account) %>% slice_head(n = 10)
top_bt看 <- cent %>% arrange(desc(betweenness), account) %>% slice_head(n = 10)
```

Interpretation. The seed, sifu.tweety.fish, ranks first by in-degree (18) and first by betweenness (69,077). Its in-degree means that 18 of its 59 friends follow it back. Part of its centrality comes from the sampling design, because every account in the network was reached through the seed. Apart from the seed, the most followed accounts are regimecpa.bsky.social, philpax.me, danioloc.xyz, segyges.bsky.social, theophite.bsky.social (in-degree 7, 7, 6, 6, 6). These are accounts that several of the seed’s friends independently chose to follow, which makes them shared reference points in this community. Of the 8 accounts in the top-5 table, 8 are individuals according to their public profiles, and none are news outlets or companies. They are mainly software, AI and academic voices. Bluesky’s chief technology officer (pfrazee.com) is also among the most followed accounts, with an in-degree of 5. After the seed, every account in the betweenness top 10 is one of the seed’s friends.

Table 4: Top 10 accounts by betweenness

account	role	in-degree	betweenness	posted?
sifu.tweety.fish	Seed	18	69,077	TRUE
regimecpa.bsky.social	Friend	7	6,593	FALSE
emilymoin.com	Friend	2	4,712	FALSE
jefferyharrell.bsky.social	Friend	3	4,696	FALSE
mjcrockett.bsky.social	Friend	2	4,069	FALSE
denam.bsky.social	Friend	3	2,519	FALSE
jmiers230.bsky.social	Friend	2	2,466	FALSE
emilyrcwilson.bsky.social	Friend	4	2,359	FALSE
bill-of-lefts.bsky.social	Friend	2	2,336	FALSE
joshuagraha12.bsky.social	Friend	1	2,304	FALSE

Table 5: Who are the most central accounts? (top 5 by each measure)

account	role	in-deg.	betw.	type	profile says
sifu.tweety.fish	Seed	18	69,077	Individual	Ex-cDc hacker, ML PhD, writer
regimecpa.bsky.social	Friend	7	6,593	Individual	Accountant, satire
philpax.me	Friend of friend	7	0	Individual	Software engineer
daniiloc.xyz	Friend of friend	6	0	Individual	Technologist
segyges.bsky.social	Friend of friend	6	0	Individual	AI newsletter writer
emilymoin.com	Friend	2	4,712	Individual	Physician, ethicist
jefferyharrell.bsky.social	Friend	3	4,696	Individual	Tech hobbyist
mjcrockett.bsky.social	Friend	2	4,069	Individual	Academic (psychology)

Betweenness leaders. regimecpa.bsky.social is the only account apart from the seed that ranks in the top five on both measures, so it is both widely followed and a bridge. Among the remaining accounts, the highest betweenness belongs to emilymoin.com (betweenness 4,712, in-degree 2, follows 96 sampled accounts, 60 of which no other sampled account follows); jefferyharrell.bsky.social (betweenness 4,696, in-degree 3, follows 50 sampled accounts, 36 of which no other sampled account follows); mjcrockett.bsky.social (betweenness 4,069, in-degree 2, follows 50 sampled accounts, 40 of which no other sampled account follows). These accounts have low in-degree, so few accounts in the sample follow them, and they are not popular in the in-degree sense. Their betweenness comes from their position instead. Each one is the only route from the seed to a large set of accounts that no other sampled account follows, so removing any of them would cut those accounts off from the rest of the network. By their public profiles they are: physician, ethicist; tech hobbyist; academic (psychology). They therefore connect the seed’s circle to audiences outside security rather than to security communities.

Betweenness must be read with care here. Only the seed and its friends have outgoing edges, so only these 60 accounts can lie on a path between other accounts. Betweenness therefore ranks bridges *within the sample*, not across Bluesky as a whole.

Visualising the network

The figure above shows the largest weakly connected component. Because the network has only one component, this is the whole network. It has a star-like structure: the seed sits at the centre, and each friend has its own “fan” of followed accounts. Fans overlap only where friends follow the same accounts. With 3,201 nodes, the graph is too dense to read account names, so we also plot a subgraph.

Subgraph choice. Following the project guide, we take the **50 accounts with the highest total degree** in the largest weakly connected component, together with the edges among them. At 50 nodes the subgraph is small enough to label. Total degree here mostly measures how many follows we *collected* for an account, so this subgraph highlights the sampled follow activity around the seed. It does not necessarily show the community’s most influential accounts. Those are better identified by in-degree (see the in-degree table above).

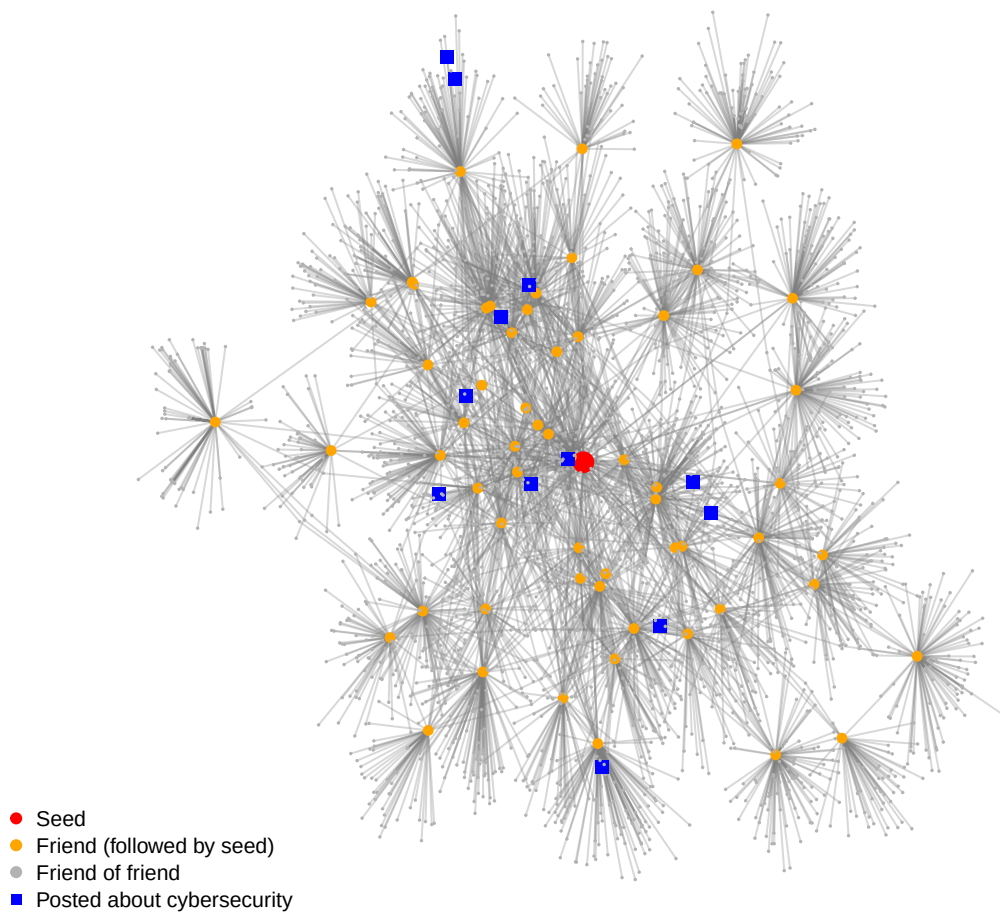


Figure 3: Largest weakly connected component of the follow network (here, the full network). Red = seed, orange = the seed's friends, grey = friends of friends, blue squares = other accounts that posted about cybersecurity in our data

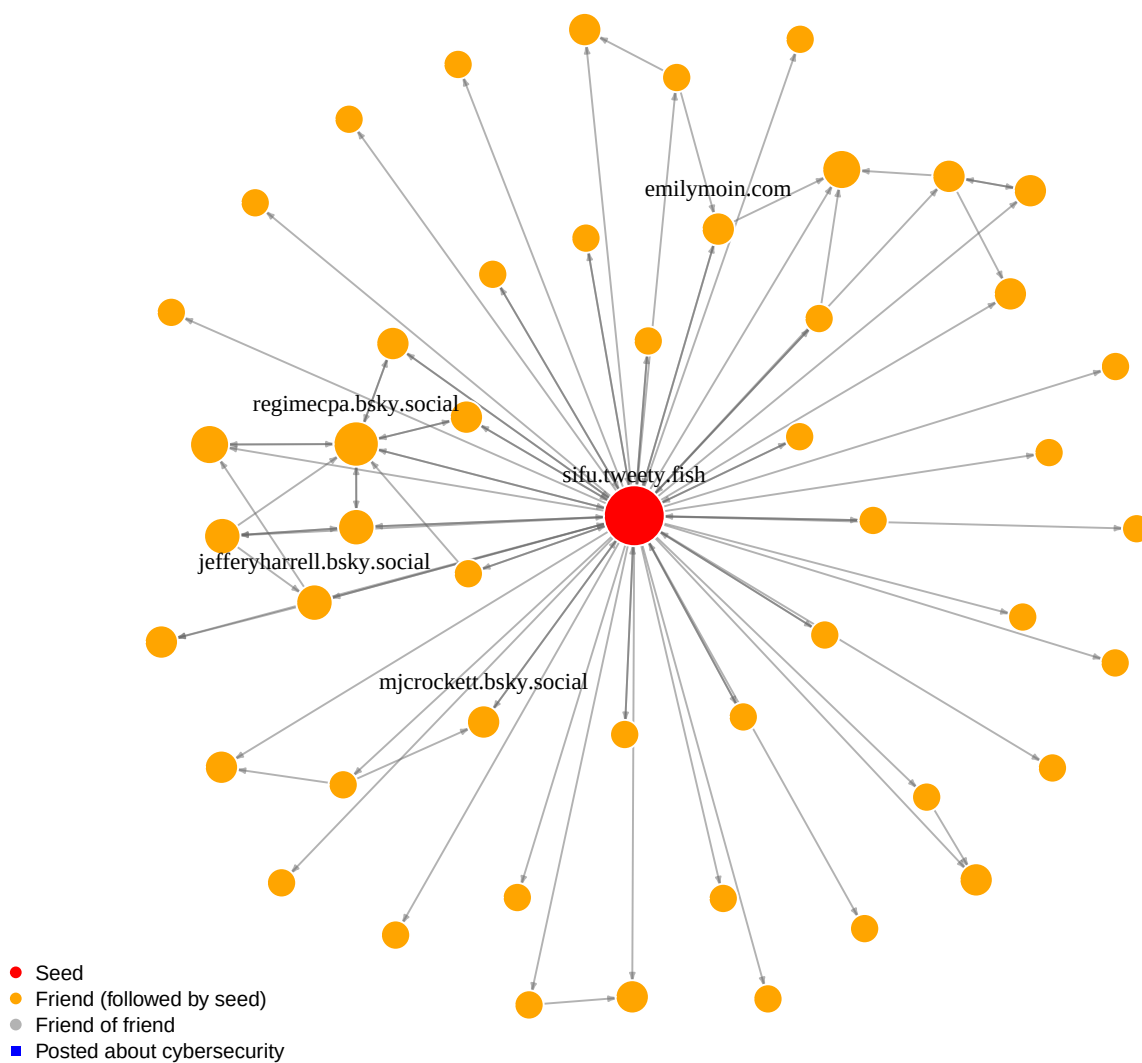


Figure 4: Subgraph of the 50 accounts with the highest total degree in the largest weakly connected component. Node size = in-degree within the full network; red = seed, orange = the seed's friends

Table 6: Cybersecurity posts by authors inside vs outside the follow network

in_network	authors	posts	median_likes	mean_likes
FALSE	561	1273	0.0	1.8
TRUE	13	20	8.5	29.1

Table 7: Most common Section 2.4 cluster for network accounts that posted about cybersecurity

account	main_cluster	posts_in_cluster
bdowney.bsky.social	5	4
davidho.bsky.social	4	1
dmnd.me	4	1
doctorow.pluralistic.net	4	1
hikikomorphism.bsky.social	5	1
k8em0.bsky.social	5	1
mattweiner19.bsky.social	4	1
persistentpoasting.bsky.social	5	1
sifu.tweety.fish	5	3
timkellogg.me	5	1
volts.wtf	5	1
vortexegg.com	5	1

The subgraph contains the seed, 49 of its friends and 0 friends of friends, joined by 93 edges. Total degree is dominated by the follow lists we collected, so the 50 highest-degree accounts are all accounts whose follows were sampled, and the commonly followed hubs identified by in-degree do not appear here. Of the subgraph’s 93 edges, 49 run from the seed to its friends and 17 run from friends back to the seed. There are only 36 follow edges among the 59 friends themselves, so they are only loosely tied to one another and are held together mainly through the seed.

Link to the other analyses

```
net_authors <- intersect(V(g)$name, authors)

link_tab <- posts %>%
  mutate(in_network = author %in% net_authors) %>%
  group_by(in_network) %>%
  summarise(authors = n_distinct(author), posts = n(),
            median_likes = median(like_count), mean_likes = round(mean(like_count), 1))
ktab(link_tab, caption = "Cybersecurity posts by authors inside vs outside the follow network")
```

Only 13 of the 3,201 accounts in the network (0.4%) wrote any of the 1,293 cybersecurity posts in our sample. The seed’s profile describes a former member of the hacker group cDc and an ML researcher, but the accounts it and its friends follow are mostly general technology, AI and academic voices rather than a cybersecurity community. Posts from these 13 network members receive more engagement than other posts (median 8.5 vs 0 likes). These are only 20 posts and include the seed’s own widely liked posts, so this is a descriptive comparison, not a test. It is consistent with Section 2.2, where human accounts were far more likely to be engaged with than bots.

The table above links the network to the clustering in Section 2.4 by giving each posting network account its most common cluster. 2 different cluster(s) appear among these 12 accounts. The guide suggested colouring the network nodes by cluster, but cluster labels exist only for accounts that posted, which is 13 of 3,201 nodes. Colouring the plots by cluster would leave more than 99% of nodes uncoloured, so we report this link as a table instead. This is a documented deviation from the guide.

Answer to the research question

- **Most central accounts.** The seed, sifu.tweety.fish, is the most central account on both measures, partly by design. Among the other accounts, popularity (in-degree) is held by accounts such as regimecpa.bsky.social, philpax.me, daniloc.xyz, which several of the seed’s friends follow. Bridging (betweenness) is held by friends of the seed such as emilymoin.com, jefferyharrell.bsky.social, mjcrockett.bsky.social.
- **Type of accounts.** 8 of the 8 most central accounts we reviewed are individuals, and none are news outlets or companies. Most work in software, AI or academia.
- **Is it a cybersecurity community?** No. Only 13 of 3,201 accounts (0.4%) posted about cybersecurity in our sample. The seed has a security background, but the community it follows is a general technology and AI audience. Within this sample, cybersecurity discussion on Bluesky is not concentrated in this highly followed human’s network.

Limitations

- **Follow network instead of mention network.** Mentions were too rare to build a mention network, so this analysis shows who follows whom, not who talks to whom about cybersecurity.
- **Two-step sample around one seed.** The network covers only the seed, its 59 friends and the accounts they follow, so it is an *ego network*, not the cybersecurity community on Bluesky. A different seed would give a different network.
- **Capped follow lists.** The API returned at most 99 follows per friend, so most friends’ lists are incomplete, and some edges among sampled accounts are missing.
- **Sampling shapes centrality.** Only the seed and friends have outgoing edges. This inflates their total degree and betweenness, and it means the top-50 subgraph reflects the sample design.
- **Snapshot in time.** Follows were collected once (06 October 2026), and follow relationships change over time.
- **Account types** come from reading public profile descriptions, which are self-reported and can change.

3. Overall Findings

This section answers the four research questions and then shows how the answers fit together. Engagement means at least one like, repost or reply, measured by the variables `like_count`, `repost_count` and `reply_count` at collection time.

RQ1: Text analysis (Section 2.3). *What topics and words dominate cybersecurity posts on Bluesky?* Using the word counts from Section 2.3, the ten most frequent words are post (407), ai (346), ransomware (284), blog (212), ransom (192), title (192), cti (191), threatintelligence (190), sector (152), security (145). The 20 most frequent words point to ransomware and breaches, vulnerabilities, AI and threat intelligence among the main themes. Several frequent words come mainly from the ten most active accounts rather than from the wider discussion. More than half of all uses of “post”, “ransomware”, “blog”, “ransom”, “title” appear in their posts, so word frequencies are shaped strongly by a few high-volume accounts.

RQ2: Clustering analysis (Section 2.4). *What groups of similar cybersecurity posts can be identified?*

The clustering in Section 2.4 grouped the posts into 6 clusters: **5** (780 posts, 62%; top words: security, data, news); **4** (191 posts, 15%; top words: ai, security, openai); **6** (190 posts, 15%; top words: post, blog, cti); **1** (52 posts, 4%; top words: cve, chromium, confusing); **3** (29 posts, 2%; top words: ransomware, cyberangriff, teaminfosec); **2** (19 posts, 2%; top words: job, jobs, educativ.net). The largest cluster holds 62% of posts.

RQ3: Hypothesis test (Section 2.2). *Are posts from human accounts more likely to receive engagement than posts from automated (bot) accounts?* Yes, in this sample. A chi-squared test of independence rejected the null hypothesis that account type and engagement are independent ($\chi^2 = 547.67$, $df = 1$, $p < 0.001$). 56.5% of human-authored posts received at least one like, repost or reply, compared with 14.5% of bot posts, so human-authored posts had an engagement rate about 3.9 times higher. Bots wrote 53% of the 2,845 English posts collected. This is an association, not a causal effect, because the test does not control for follower counts or posting time.

RQ4: Network analysis (Section 2.5). *In the follow network around the most-followed active author in the discussion, which accounts are most central, and is it a cybersecurity community?* The seed, sifu.tweety.fish, is the most central account, and 18 of its 59 friends follow it back. Apart from the seed, the most central accounts we reviewed are individuals working in software, AI and academia rather than news outlets or companies. The sampled follow network does not appear to be predominantly cybersecurity-focused. Only 13 of its 3,201 accounts (0.4%) posted about cybersecurity in our sample. Because mentions were too rare to build the mention network suggested in the guide, this answer is about follow relationships.

How the analyses connect

Taken together, the four analyses describe a cybersecurity discussion on Bluesky that is high-volume but substantially automated. Section 2.1 found that bots wrote 53% of the English posts. Even after they were removed, the ten most active accounts, mostly alert and news feeds, still wrote 32% of the cleaned posts. The text analysis shows the same pattern, since 11 of the 20 most frequent words come mainly from these accounts. In the clustering, cluster 3 has the highest share of posts from these accounts (100%) and cluster 2 the lowest (0%), so the clusters partly separate feed-style content from more conversational posts. The hypothesis test shows that bot posts attracted less engagement: human-authored posts were about 3.9 times as likely to be engaged with. The network analysis adds that the most-followed active author we sampled sits in a general technology and AI community. Its network members wrote only 20 of the 1,293 posts, but those posts had a median of 8.5 likes compared with 0 for other posts. **Main conclusion:** the results suggest that, in this sample, automated accounts contributed substantially to the volume of cybersecurity posts on Bluesky. Human-authored posts were more likely to receive engagement, and the most-followed human voice we sampled belongs to a network that extends beyond the security field.

Overall limitations

- **Short, keyword-based sample.** The data cover about five days of posts matching “cybersecurity” or “infosec”. The results reflect that week’s news, and posts on the topic that use neither keyword are missed.
- **English only.** Posts not tagged as English were removed, and some posts in other languages carry an English tag.
- **Imperfect bot detection.** Bots were identified by handle and manual review. Automated feeds without “bot” in their name remain in the cleaned data and affect the word counts and engagement results.
- **Engagement at collection time.** Newer posts had less time to gather likes, reposts and replies.
- **One-seed follow network.** The network is a two-step sample around a single account with capped follow lists. It shows who follows whom, not who talks to whom.

Code Repository and Reproducibility

GitHub repository: Group 32 — COMP3020 project

The repository is intended to provide the R Markdown source files, analysis scripts, and saved datasets used for the project. The data collection and cleaning scripts are referenced as `scripts/Collect.R` and `scripts/Clean.R`. The report uses saved datasets rather than collecting new data when knitted. **Before submission, confirm the link opens without signing in and that the referenced files are present.**